Google Colab Python Code Handbook

Complete Python Scripts Collection for PDF Data Upload, Extraction, Cleaning, Analysis & Visualization

Is handbook me Google Colab me PDF sales dataset load karne, missing & incorrect values clean karne, total summary stats calculate karne, aur visual charts banane ke saare Python codes ek jagah structured format me hain.

1. Google Colab me File Upload karne ka Code:
Local computer se PDF file upload karne ke liye simple code snippet.

```
from google.colab import files
uploaded = files.upload()
```

2. Uploaded PDF ka Text Dekhne & Files List karne ka Code:
Colab directory check karne aur PDF file ka text extract karne ka snippet.

```
# A. Uploaded PDF se text extract karne ke liye
!pip install pypdf
from pypdf import PdfReader

reader = PdfReader('Electronics_Sales_Dirty_Dataset_200.pdf')
print(reader.pages[0].extract_text())

# B. Colab directory me files dekhne ke liye
import os
print(os.listdir())
```

3. Basic PDF Table Extraction & Data Cleaning Code:
PDF se tables extract karke missing values, extra spaces, aur duplicates saaf karne ka script.

```
!pip install pdfplumber pandas

import pdfplumber
import pandas as pd

# 1. PDF se tables extract karke DataFrame banayein
data = []
with pdfplumber.open('Electronics_Sales_Dirty_Dataset_200.pdf') as pdf:
    for page in pdf.pages:
        table = page.extract_table()
        if table:
            data.extend(table)

# Pehli row ko column headers banayein
df = pd.DataFrame(data[1:], columns=data[0])

# 2. Automatic Data Cleaning
# A. Unwanted spaces hatayein
df = df.apply(lambda x: x.str.strip() if x.dtype == "object" else x)

# B. Duplicate rows hatayein
df = df.drop_duplicates()

# C. Missing values fill karein
df = df.fillna('Unknown')

# 3. Cleaned Data ko CSV me save karein
print("Cleaned Data Shape:", df.shape)
df.to_csv('Cleaned_Electronics_Sales.csv', index=False)
```

4. Advanced Cleaning & Cleaned Data Ki PDF Report Banane Ka Code:

City names, payment methods, fraud labels normalize karke final PDF export karne ka script.

```
!pip install pdfplumber pandas reportlab

import pdfplumber
import pandas as pd
from reportlab.lib.pagesizes import A4, landscape
from reportlab.platypus import SimpleDocTemplate, Table, TableStyle, Paragraph, Spacer
from reportlab.lib.styles import getSampleStyleSheet
from reportlab.lib import colors

# 1. PDF se data extract karke DataFrame banayein
data = []
with pdfplumber.open('Electronics_Sales_Dirty_Dataset_200.pdf') as pdf:
    for page in pdf.pages:
        table = page.extract_table()
        if table:
            data.extend(table)

df = pd.DataFrame(data[1:], columns=data[0])

# 2. Data Cleaning
df = df.apply(lambda x: x.str.strip() if x.dtype == "object" else x)

for col in ['City', 'Location', 'Store_Location']:
    if col in df.columns:
        df[col] = df[col].str.title()

for col in ['Payment', 'Payment_Method', 'Status']:
    if col in df.columns:
        df[col] = df[col].str.upper()

df = df.fillna('N/A')
df = df.drop_duplicates()

# 3. Cleaned Data ki PDF File Banayein
pdf_filename = "Cleaned_Electronics_Sales_Dataset.pdf"
doc = SimpleDocTemplate(pdf_filename, pagesize=landscape(A4))
elements = []

styles = getSampleStyleSheet()
elements.append(Paragraph("<b>Cleaned Electronics Sales Dataset Report</b>", styles['Title']))
elements.append(Spacer(1, 12))

table_data = [df.columns.tolist()] + df.values.tolist()
t = Table(table_data)
t.setStyle(TableStyle([
    ('BACKGROUND', (0, 0), (-1, 0), colors.navy),
    ('TEXTCOLOR', (0, 0), (-1, 0), colors.whitesmoke),
    ('ALIGN', (0, 0), (-1, -1), 'CENTER'),
    ('FONTNAME', (0, 0), (-1, 0), 'Helvetica-Bold'),
    ('FONTSIZE', (0, 0), (-1, -1), 8),
    ('BOTTOMPADDING', (0, 0), (-1, 0), 6),
    ('GRID', (0, 0), (-1, -1), 0.5, colors.grey),
]))

elements.append(t)
doc.build(elements)
print(f"Cleaning complete! Saved as: {pdf_filename}")
```

5. Cleaned Data Ki Total Sales Summary Calculation Code:

Overall Revenue, Total Units, Avg Rating, City Sales, Category Performance, aur Payment Stats calculate karne ka code.

```
import pandas as pd

# Cleaned CSV file load karein
df = pd.read_csv('Cleaned_Electronics_Sales.csv')

# Numeric columns convert karein
df['Total Amount'] = pd.to_numeric(df['Total Amount'].astype(str).str.replace('■', '').str.replace(',', ''), er
df['Qty'] = pd.to_numeric(df['Qty'], errors='coerce')
df['Rating'] = pd.to_numeric(df['Rating'], errors='coerce')

# 1. Overall Key Summary Metrics
print("===== ■ OVERALL DATASET SUMMARY =====")
print(f"Total Transactions: {len(df)}")
print(f"Total Revenue (Sales): ■{df['Total Amount'].sum():,.2f}")
print(f"Total Quantity Sold: {df['Qty'].sum():,.0f}")
print(f"Average Customer Rating: {df['Rating'].mean():.2f} / 5.0")

# 2. City / Branch Performance Summary
print("\n===== ■ CITY-WISE SALES SUMMARY =====")
city_col = 'City Branch' if 'City Branch' in df.columns else 'City'
city_summary = df.groupby(city_col)['Total Amount'].sum().reset_index()
print(city_summary.sort_values(by='Total Amount', ascending=False).to_string(index=False))

# 3. Top Selling Product Categories
print("\n===== ■ CATEGORY-WISE SALES SUMMARY =====")
category_summary = df.groupby('Product Category')['Total Amount'].sum().reset_index()
print(category_summary.sort_values(by='Total Amount', ascending=False).to_string(index=False))

# 4. Payment Method Summary
print("\n===== ■ PAYMENT METHOD SUMMARY =====")
payment_summary = df['Payment Method'].value_counts().reset_index()
payment_summary.columns = ['Payment Method', 'Count']
print(payment_summary.to_string(index=False))
```

6. Summary Statistics Visual Charts (Bar & Pie Charts) Code:

City-wise sales, Category revenue, Payment share, aur Quantity sold ke 4 visual charts generate karne ka script.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Cleaned CSV file load karein
df = pd.read_csv('Cleaned_Electronics_Sales.csv')

# Numeric columns convert karein
df['Total Amount'] = pd.to_numeric(df['Total Amount'].astype(str).str.replace('■', '').str.replace(',', ''), er
df['Qty'] = pd.to_numeric(df['Qty'], errors='coerce')

# Visual style set karein
sns.set_theme(style="whitegrid")
fig, axes = plt.subplots(2, 2, figsize=(14, 10))

# 1. Bar Chart: City-Wise Total Sales
city_col = 'City Branch' if 'City Branch' in df.columns else 'City'
city_sales = df.groupby(city_col)['Total Amount'].sum().reset_index()
sns.barplot(data=city_sales, x=city_col, y='Total Amount', ax=axes[0, 0], palette='Blues_r')
axes[0, 0].set_title('City-Wise Total Sales (Revenue)', fontsize=12, fontweight='bold')
axes[0, 0].set_xlabel('City')
axes[0, 0].set_ylabel('Total Sales (■)')

# 2. Bar Chart: Category-Wise Sales
cat_sales = df.groupby('Product Category')['Total Amount'].sum().reset_index().sort_values(by='Total Amount', as
sns.barplot(data=cat_sales, x='Product Category', y='Total Amount', ax=axes[0, 1], palette='Greens_r')
axes[0, 1].set_title('Sales by Product Category', fontsize=12, fontweight='bold')
axes[0, 1].set_xlabel('Category')
axes[0, 1].set_ylabel('Total Sales (■)')
axes[0, 1].tick_params(axis='x', rotation=30)

# 3. Pie Chart: Payment Method Distribution
payment_counts = df['Payment Method'].value_counts()
axes[1, 0].pie(payment_counts, labels=payment_counts.index, autopct='%1.1f%%', startangle=140, colors=sns.color_
axes[1, 0].set_title('Payment Method Share', fontsize=12, fontweight='bold')

# 4. Bar Chart: Total Quantity Sold by Category
cat_qty = df.groupby('Product Category')['Qty'].sum().reset_index().sort_values(by='Qty', ascending=False)
sns.barplot(data=cat_qty, x='Product Category', y='Qty', ax=axes[1, 1], palette='Oranges_r')
axes[1, 1].set_title('Total Units Sold by Category', fontsize=12, fontweight='bold')
axes[1, 1].set_xlabel('Category')
axes[1, 1].set_ylabel('Units Sold')
axes[1, 1].tick_params(axis='x', rotation=30)

plt.tight_layout()
plt.show()
```